Ex:-12:- [B.Q:-2] :- The % profit when an article is sold for 400/- is twice the loss when the same article is sold for 250/-. The C.P is ?

$P = 2L \Rightarrow [400 - C.P] = 2[CP - 250]$

$\Rightarrow 400 - CP = 2CP - 500 \Rightarrow 3CP = 900$

$\Rightarrow \boxed{CP = 300}$

Ex:-13:- [B.Q:-23] :- A shopkeeper bought 120 chairs @ 250 each. He spends 3000 rupees on transportation. He marked 400/- each & gives 10% discount. The profit/loss % is ?

Ans:- C.P of 120 chairs = $120 \times 250 + 3000 = 33000$.

S.P of 120 chairs = $120 \times (400 \times 0.9$ (or) $400 - 40)$

$= 120 \times 360 = 43200$

$P\% = \frac{43200 - 33000}{33000} \times 100 = \frac{340}{11}\% = 30\frac{10}{11}\%$.

Ex:-14:- [B.Q:-24:-] :- The M.R.P of an article is 50% above C.P. When the Marked Price is fed by 20% & S.P is fed by 20%, then the profit doubles. If the marked price (mrp) is 3000. Then original selling Price is ?

Ans:- $M.P_1 = (100\% + 50\%)$ of $CP = 3000 \Rightarrow CP = 2000$

If $\Rightarrow M.P_2 = 1.2 \times MP_1 = 1.2 \times 3000 = 3600$, then $SP_2 = 1.2\ SP_1$

then Profit doubles $\Rightarrow 1.2SP_1 - CP = 2(SP_1 - CP)$

$\Rightarrow 2SP_1 - 1.2SP_1 = CP$

$\Rightarrow 0.8\ SP_1 = CP = 2000$

$SP_1 = \frac{2000}{0.8} = \frac{20000}{8}$

$\boxed{SP_1 = \text{Original S.P} = 2500}$

Ex:-15:-[B.Q:25]:-

A man sells a DVD player at a profit of 20%. Had he bought it at 25% less & sold it for 840/- less, he would have gained 10%. What is the cost price of the DVD player?

Ans:- $CP_1 = 100\%\text{ of }CP_1 \Rightarrow SP_1 = 120\%\text{ of }CP_1 = 1.2CP_1$

$CP_2 = 0.75\,CP_1$ [75% of CP_1] $\Rightarrow SP_2 = 1.2CP_1 - 840$

$\Rightarrow 1.2CP_1 - 840 = 1.1(0.75P_1)$

$\Rightarrow 1.2CP_1 - 0.825\,CP_1 = 840$

$\Rightarrow 0.375\,CP_1 = 840$

$\Rightarrow CP_1 = \text{Original C.P} = \frac{840 \times 1000}{375}$

Cost Price = 2240/-.